

A SUPPLEMENTARY FINITE-PLACE RATIONAL CONTROL

THE CLANKERS

This is an additional source fragment preserved from the derivation of this tranche. The main manuscript contains the simpler joint-channel one-prime construction; neither statement is a counterexample to Erdős–Straus.

1. A FINITE-PLACE AND BOUNDED-SIZE TEST DOES NOT ENSURE INTEGRALITY

The supplied real separation estimates [5, §1.2, equations (2)–(4)] are necessary at an integral witness. The next theorem checks their logical strength on explicit targets. It is not a new assertion about the Brauer group. For the distinction between natural solutions and arithmetic approximation see [2, Theorems 1.1, 1.8 and 1.9].

Theorem 1.1 (Explicit rational controls for any fixed finite place set). *Let Σ be any fixed finite set of rational primes. There are infinitely many primes $p \equiv 1 \pmod{840}$ with a positive rational triple (x, y, z) such that the following all hold:*

- (1) $4/p = 1/x + 1/y + 1/z$, with $p/4 < x < p/2$ and four separated literal roots;
- (2) $x, y, z \in \mathbb{Z}_q$ for every $q \in \Sigma \cup \{p\}$;
- (3) the literal prime-local completion has precisely the Type-II signatures, Smith exponents, conductor and four-rational-point case proved above;
- (4) with $s_p = (p^2 + 3p)/4$, $B_p = p + s_p(s_p + 1)$,
 $S < B_p$, $\max(x, y, z) < s_p(s_p + 1)/3$, $\text{Disc}(\mathcal{H}) \geq 144/B_p^6$, $2/S \leq |\mathcal{H}'(\ell_i)| \leq S^2$;
- (5) one specified prime $l \notin \Sigma \cup \{p\}$ has $v_l(z) = -1$, so this particular triple is not integral;
- (6) the same p has a separately displayed original integer E solution.

Proof. Choose a prime $l \geq 13$ outside Σ , and let $M = l^2 \prod_{q \in \Sigma} q$. Use the reduced progression

$$(IC32) \quad p \equiv 1 \pmod{840}, \quad p \equiv 1 \pmod{l}, \quad p \equiv -1 \pmod{11}.$$

Its moduli are pairwise coprime, and its residues are units. Dirichlet’s theorem [4, Theorem 18.1] supplies infinitely many prime members. Take $p > \max \Sigma$ and $p^2 \geq 128M$, and set $a = (p + 3)/4$. Select b by CRT:

$$(IC33) \quad b \equiv 3 \pmod{p}, \quad b \equiv \begin{cases} 1 & \pmod{q}, \quad q \mid a, \\ 0 & \pmod{q}, \quad q \nmid a, \end{cases} \quad (q \in \Sigma), \quad 3b - a \equiv l \pmod{l^2}.$$

There is one residue modulo pM . Add pM to its least nonnegative representative, so $pM \leq b < 2pM$. Put

$$(IC34) \quad c = \frac{ab}{3b - a}, \quad (x, y, z) = (a, pb, pc).$$

Then $a/3 < c < a/2$, and direct addition proves the ES identity. For $q \in \Sigma$, $3b - a$ is a unit: if $q \mid a$ then $q \neq 3$ and $b = 1$; otherwise use $b = 0$. At l , $a \equiv 1$, so $b \equiv 1/3$ and $v_l(3b - a) = 1$, while ab is a unit. Thus $v_l(z) = -1$, with no cancellation.

At p , the normalized cluster roots are $1, 3, 3/11$. The character of 11 is -1 by (IC32) and reciprocity. The derivative residues in (IC14) consequently have signs $(-1, -1, 1)$, because $3, 5, -1$ are squares at a hard prime. All distinctness and unit conditions used in the cluster and normalization theorems (Theorems 2.2 and 4.1) of the companion workbook proof therefore hold literally for this rational triple, giving the four-point M case. No global square-divisor state is inferred.

For the quantitative bounds, $p < a/3 \cdot p < pc < pa/2 < pb$ and $a < p$; more precisely $pc - p > p(p - 9)/12 > 1$ for $p \geq 13$. The other two successive gaps in the ordered list a, p, pc, pb are also greater than one. Further,

$$S < (2M + 1)p^2 \leq 3Mp^2, \quad \max(x, y, z) < 2Mp^2.$$

Since $B_p \geq p^4/16$ and $s_p(s_p + 1)/3 \geq p^4/48$, the condition $p^2 \geq 128M$ proves the two stated upper bounds. Four real numbers with successive gaps at least one have Vandermonde modulus at least $1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = 12$. At a selected root the product of its three difference moduli is at least two and at most S^3 . These prove the displayed discriminant and derivative bounds.

Finally $11 \mid p + 1$. The separately original state

$$R_0 = 11, \quad a_0 = u_0 = (p + 11)/4, \quad (h, r, s, \kappa) = (a_0, 1, 1, (p + 1)/11)$$

returns $(a_0, a_0\kappa, pa_0\kappa)$. This proves the last assertion and excludes any interpretation of the construction as an ES counterexample. $\square$

For $\Sigma = \{2, 3, 5, 7, 11\}$ and $l = 13$, the first certified example used here is

$$(IC35) \quad p = 207481, \quad a = 51871, \quad b = 95637744510, \quad c = \frac{4960825445478210}{286913181659}.$$

The nonintegral denominator in the rational ES triple is

$$z = \frac{1029277024253264489010}{286913181659}, \quad v_{13}(z) = -1.$$

Its independent original endpoint solution has denominators

$$(51873, 978428526, 203005329003006).$$

The certificate also checks prime parameters 567841 and 687961 in the same progression. Trial division proves each of these primality assertions.

Theorem 1.1 does not concern all places simultaneously, nor a p -dependent complete set of possible denominator primes. The omitted prime is an explicit place where the target fails the original integrality condition. It shows why a fixed finite set of local checks, even with these exact real bounds and the rank-eight completion, is not itself that condition.

REFERENCES

- [1] C. Elsholtz and T. Tao (2013), *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, Journal of the Australian Mathematical Society 94, 50–105. <https://arxiv.org/abs/1107.1010>, §2, Propositions 2.2 and 2.6.
- [2] M. Bright and D. Loughran (2020), *Brauer–Manin obstruction for Erdős–Straus surfaces*, Bulletin of the London Mathematical Society 52(4), 746–761. <https://arxiv.org/abs/1908.02526>, Theorem 1.1, Corollary 1.3, Theorems 1.8–1.9 and Appendix A.
- [3] J. S. Milne (2020), *Algebraic Number Theory*, version 3.08, Chapter 7, Newton’s lemma and the sections on unramified and totally ramified local extensions, pp. 119–130. <https://www.jmilne.org/math/CourseNotes/ANT.pdf>.
- [4] A. V. Sutherland (2017), *Dirichlet L-functions, primes in arithmetic progressions*, MIT 18.785 Lecture 18, Theorem 18.1. <https://math.mit.edu/classes/18.785/2017fa/LectureNotes18.pdf>.
- [5] The Clankers / Split-Zero continuation (2026), *Combined continuation: literal ES quartic, fixed-input monodromy, finite rank-eight completion and native error bounds*, supplied as `Pasted markdown(20260920-025519).md`. The original text is preserved byte-for-byte in `antecedent/attached_continuation.md`. The literal target and height formulas are in §1; the rank-two/rank-six split and deck labels are in §2; the signed completion, open inverse and multiple-root algebras are equations (18)–(24) of §3. Those constructions are antecedents, not claimed as discoveries of the present tranche.
- [6] The Clankers (2026), *ES–Fabel arithmetic bridge*, `es_turn7_fabel_bridge_20260920/core.tex`, §§2–5 and `root_orders.tex`. The reciprocal cubic and its seven-point chart are kept separate from the literal quartic used here.
- [7] The Clankers (2026), *Fixed-target capacity completion*, `es_turn7_capacity_completion_20260920/core.tex`, complete same-grade channel return, fixed-target classification, sharp finite repair, and original prime realization. The exact upstream files and hashes are listed in `source_reading.json`. No universal occupancy statement is imported.